ATM Option Approximations

Derivations and intuition for desk-level mental pricing

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Study notes

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Abstract

These notes derive from first principles the standard at-the-money approximations used by options traders for fast mental pricing: the Brenner–Subrahmanyam premium formula, the ATM expressions for vega, gamma, and theta, the put–call symmetry of vega and gamma in delta-space, and the dollar-gamma–vega identity. Every result is proved from the Black–Scholes formula. Throughout we work with zero rates and zero dividends; carry corrections do not affect the structure of the approximations.

The single observation underlying all the formulas is that the standard normal density evaluated at zero is $\phi(0) = 1/\sqrt{2\pi} \approx 0.4$, and that $d_1 \approx 0$ for at-the-money short-dated options. Once that fact is internalized, every approximation in this document falls out mechanically.

Contents

1. Setup and notation

We work in the standard Black–Scholes framework. For a European call on an underlying with spot price S, strike K, time to expiry T, volatility σ, and zero risk-free rate, the price is

$$C(S,K,T,\sigma) \;=\; S\cdot N(d_1) \;-\; K\cdot N(d_2), \tag{1}$$

$$d_1 \;=\; \frac{\ln(S/K)+\frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 \;=\; d_1-\sigma\sqrt{T}, \tag{2}$$

where N is the standard normal CDF and ϕ denotes its density:

$$\phi(x) \;=\; \frac{1}{\sqrt{2\pi}}\,e^{-x^2/2}. \tag{3}$$

The corresponding put price follows by put–call parity (with $r=0$):

$$P \;=\; C-S+K \;=\; K\cdot N(-d_2)-S\cdot N(-d_1). \tag{4}$$

We use the convention that "vega" refers to $\partial V/\partial\sigma$ (per unit change in σ, i.e., a change of 1.0, equivalent to 100 volatility points). Practitioner "vega per vol point" is this divided by 100.

2. The number 0.4

The constant 0.4 pervades every ATM approximation in this document. It comes from a single source.

Result 2.1 (Standard normal density at zero)**.**

$$\phi(0) \;=\; \frac{1}{\sqrt{2\pi}} \;=\; \frac{1}{\sqrt{6.2832\ldots}} \;=\; 0.39894\ldots \;\approx\; 0.4.$$

2.1 Why $d_1 \approx 0$ at ATM

For an at-the-money option with $S=K$, the logarithmic term in d_1 vanishes, leaving

$$d_1\big|_{S=K} \;=\; \frac{\sigma^2 T/2}{\sigma\sqrt{T}} \;=\; \frac{\sigma\sqrt{T}}{2}. \tag{5}$$

For typical short-dated options the quantity $\sigma\sqrt{T}$ is small. With $\sigma=0.15$ and $T=1/12$, we have $\sigma\sqrt{T}\approx 0.043$, so $d_1\approx 0.022$. To first order in $\sigma\sqrt{T}$, the density is

$$\phi(d_1) \;\approx\; \phi(0) \;=\; \frac{1}{\sqrt{2\pi}} \;\approx\; 0.4. \tag{6}$$

Intuition. The density ϕ peaks at zero and is smooth and slowly varying near that peak. ATM short-dated options sit essentially on the peak. This is why every ATM approximation that depends on $\phi(d_1)$ collapses to the same numerical constant.

3. ATM premium: the Brenner–Subrahmanyam formula

The most useful single approximation in mental pricing.

Result 3.1 (Brenner–Subrahmanyam). *For an at-the-money European call with zero rates,*

$$C_{ATM} \approx \frac{S\sigma\sqrt{T}}{\sqrt{2\pi}} \approx 0.4\,S\sigma\sqrt{T}.$$

Proof. At ATM with $S = K$, we have $d_1 = \sigma\sqrt{T}/2$ and $d_2 = -\sigma\sqrt{T}/2$. Use the first-order Taylor expansion of N around zero:

$$N(x) \approx N(0) + N'(0) \cdot x = \tfrac{1}{2} + \phi(0) \cdot x = \tfrac{1}{2} + \frac{x}{\sqrt{2\pi}}. \tag{7}$$

Substituting into the Black–Scholes formula with $K = S$:

$$C_{\text{ATM}} \approx S\left[\tfrac{1}{2} + \frac{\sigma\sqrt{T}/2}{\sqrt{2\pi}}\right] - S\left[\tfrac{1}{2} - \frac{\sigma\sqrt{T}/2}{\sqrt{2\pi}}\right] \tag{8}$$

$$= \frac{S\sigma\sqrt{T}}{\sqrt{2\pi}}. \tag{9}$$

The constant $S/2$ terms cancel; the linear corrections add. □ □

Remark. The straddle is twice this. At ATM with $r = 0$, put–call parity gives $C = P$, so the straddle price is

$$C + P = 2C_{\text{ATM}} \approx 0.8\,S\sigma\sqrt{T}. \tag{10}$$

3.1 Probabilistic interpretation

The formula has a clean meaning. For a normal random variable X with mean zero and standard deviation $\sigma\sqrt{T}$ (the log-return of the underlying over $[0, T]$ under the risk-neutral measure with zero drift), the expected absolute value is

$$\mathbb{E}|X| = \sigma\sqrt{T} \cdot \sqrt{\frac{2}{\pi}} \approx 0.7979\,\sigma\sqrt{T}. \tag{11}$$

The ATM straddle pays the absolute log-return of the underlying at expiry (to first order in $\sigma\sqrt{T}$), and its risk-neutral price is therefore approximately $S \cdot \mathbb{E}|X|$. The factor $\sqrt{2/\pi} \approx 0.7979$ is exactly the 0.8 in the straddle approximation. The Brenner–Subrahmanyam formula is, fundamentally, a statement about the expected absolute value of a Gaussian.

4. ATM delta

The delta of the ATM option follows from the same first-order expansion of N around zero used for the premium, and it quantifies a fact every desk knows: *the ATM call is not the 50-delta option.*

Result 4.1 (ATM delta). *For an at-the-money European call with zero rates,*

$$\Delta_{\text{ATM}} = N(d_1) \approx \tfrac{1}{2} + \phi(0)\,\frac{\sigma\sqrt{T}}{2} = 0.5 + 0.2\,\sigma\sqrt{T}.$$

Proof. At ATM, $d_1 = \sigma\sqrt{T}/2$. Apply the first-order Taylor expansion $N(x) \approx \frac{1}{2} + \phi(0)\,x$ at $x = \sigma\sqrt{T}/2$ and use $\phi(0) \approx 0.4$. □

With $\sigma = 0.20$ and $T = 0.25$ (a three-month option on a 20-vol underlying), the correction is $0.2 \cdot 0.20 \cdot 0.5 = 0.02$, giving $\Delta_{\text{ATM}} \approx 0.52$. By put–call parity ($\Delta_P = \Delta_C - 1$), the ATM put carries $|\Delta_P| \approx 0.5 - 0.2\,\sigma\sqrt{T}$: the call sits slightly *above* 50 deltas, the put slightly *below*, and the two always sum to one in absolute value.

> **Where the true 50-delta strike sits.** Setting $N(d_1) = \frac{1}{2}$ requires $d_1 = 0$, i.e. $\ln(S/K) = -\sigma^2 T/2$, so $K_{50\Delta} = S\,e^{\sigma^2 T/2} > S$: the 50-delta strike lies slightly *above* spot. This is why "ATM" requires a convention—ATM-spot, ATM-forward, or the delta-neutral strike—and why the three coincide only to first order. With zero rates spot and forward agree, but the delta-neutral straddle strike still sits a factor $e^{\sigma^2 T/2}$ above them.

5. ATM vega

5.1 The exact formula

Result 5.1 (Black–Scholes vega)**.** *For a European option (call or put) at strike K,*

$$\textit{Vega} \;=\; \frac{\partial C}{\partial \sigma} \;=\; S\,\phi(d_1)\,\sqrt{T}.$$

Proof. Differentiating $C = SN(d_1) - KN(d_2)$ with respect to σ:

$$\text{Vega} \;=\; S\phi(d_1)\frac{\partial d_1}{\partial \sigma} \;-\; K\phi(d_2)\frac{\partial d_2}{\partial \sigma}. \tag{12}$$

We use the identity (valid for all strikes, $r = 0$)

$$S\,\phi(d_1) \;=\; K\,\phi(d_2), \tag{13}$$

which is established by direct algebra: substitute $d_2 = d_1 - \sigma\sqrt{T}$ into $\phi(d_2) = e^{-d_2^2/2}/\sqrt{2\pi}$, expand $d_2^2/2 = d_1^2/2 - d_1\sigma\sqrt{T} + \sigma^2 T/2$, and use $d_1\sigma\sqrt{T} - \sigma^2 T/2 = \ln(S/K)$ to obtain $\phi(d_2)/\phi(d_1) = S/K$. Combined with $\partial d_1/\partial\sigma - \partial d_2/\partial\sigma = \sqrt{T}$, the difference of the two terms collapses to

$$\text{Vega} \;=\; S\,\phi(d_1)\,\sqrt{T}. \tag{14}$$

□ □

5.2 ATM approximation

> At ATM, with $\phi(d_1) \approx 1/\sqrt{2\pi} \approx 0.4$:
>
> $$\text{Vega}_{\text{ATM}} \;\approx\; 0.4\,S\sqrt{T} \quad \text{(per unit } \Delta\sigma = 1.0\text{)},$$
>
> $$\text{Vega}_{\text{per vol pt}} \;\approx\; 0.004\,S\sqrt{T} \quad \text{(per 1\% change in } \sigma\text{)}.$$

5.3 Put–call symmetry of vega

Result 5.2 (Same-strike symmetry)**.** *For European options at the same strike K, call vega equals put vega, exactly.*

Proof. Put–call parity (with $r = 0$) gives $C - P = S - K$. The right-hand side is independent of σ. Differentiating both sides:

$$\frac{\partial C}{\partial \sigma} - \frac{\partial P}{\partial \sigma} = 0 \implies \text{Vega}_C = \text{Vega}_P = S\,\phi(d_1)\,\sqrt{T}. \tag{15}$$

□ □

5.4 Symmetric-delta vega equivalence

The case relevant to risk-reversal trading: a 25-delta call and a 25-delta put at *different* strikes.

If $\Delta_C = N(d_1) = 0.25$, then $d_1^{\text{call}} = N^{-1}(0.25) \approx -0.6745$. Delta is not the same function of d_1 for calls and puts, and this is the source of most confusion here. For a call, $\Delta_C = N(d_1)$ directly, so

$$\Delta_C = 0.25 \implies d_1^{\text{call}} = N^{-1}(0.25) \approx -0.6745.$$

For a put, $\Delta_P = N(d_1) - 1$, which is negative. A 25-delta put means $|\Delta_P| = 0.25$, i.e. $1 - N(d_1^{\text{put}}) = 0.25$, so

$$N(d_1^{\text{put}}) = 0.75 \implies d_1^{\text{put}} = N^{-1}(0.75) \approx +0.6745.$$

The value 0.75—not 0.25—appears precisely because the put delta carries the -1: to make $|\Delta_P| = 0.25$ we need $N(d_1) = 0.75$, and one checks $\Delta_P = 0.75 - 1 = -0.25$. Because the normal distribution is symmetric about zero, $N^{-1}(0.75) = -N^{-1}(0.25)$, and therefore

$$d_1^{\text{put}} = -\,d_1^{\text{call}} \approx +0.6745.$$

The two d_1 values are exact opposites; the only approximation is the rounding of $0.674489\ldots$. The two strikes are different, but:

$$\phi(-0.6745) = \phi(+0.6745) \approx 0.319, \tag{16}$$

because ϕ is symmetric: $\phi(-x) = \phi(x)$. Therefore, using (15) and the symmetry of ϕ:

$$\text{Vega}_{25\Delta C} = S\phi(d_1^{\text{call}})\sqrt{T} = S\phi(d_1^{\text{put}})\sqrt{T} = \text{Vega}_{25\Delta P}. \tag{17}$$

This equality is exact only on a symmetric smile. With a real downside skew (puts richer than equidistant calls), the 25-delta put trades at a higher implied vol than the 25-delta call. The vega expression in absolute terms then differs slightly. Consequently a delta-hedged 25/25 risk reversal is approximately, not exactly, vega-neutral; the small residual is the skew premium itself, which is the quantity skew-trading strategies attempt to harvest.

6. ATM gamma

Result 6.1 (Black–Scholes gamma). *For a European call (and put, by parity) at strike K,*

$$\Gamma \;=\; \frac{\phi(d_1)}{S\sigma\sqrt{T}}.$$

Proof. Gamma is $\partial^2 C/\partial S^2 = \partial\Delta/\partial S$. From $\Delta_{\text{call}} = N(d_1)$:

$$\Gamma \;=\; \phi(d_1)\cdot\frac{\partial d_1}{\partial S} \;=\; \phi(d_1)\cdot\frac{1}{S\sigma\sqrt{T}}. \tag{18}$$

□ □

At ATM:

$$\Gamma_{\text{ATM}} \;\approx\; \frac{1}{\sqrt{2\pi}\,S\sigma\sqrt{T}} \;\approx\; \frac{0.4}{S\sigma\sqrt{T}}.$$

6.1 Put–call gamma equality

By the same put–call parity argument as for vega, taking the second derivative of $C - P = S - K$ with respect to S:

$$\Gamma_C - \Gamma_P \;=\; \frac{\partial^2}{\partial S^2}(S-K) \;=\; 0. \tag{19}$$

At any single strike, call gamma and put gamma are identical. The symmetric-delta argument from the vega section transfers verbatim: the 25-delta call and 25-delta put have approximately equal gamma because $\phi(d_1)$ takes the same value at ± 0.6745 and the denominator $S\sigma\sqrt{T}$ is identical.

7. The vega–gamma identity (dollar gamma)

Result 7.1 (Dollar-gamma–vega identity). *For any European option,*

$$\textit{Vega} \;=\; S^2\,\sigma\,T\cdot\Gamma.$$

Proof. Direct algebra:

$$S^2\sigma T\cdot\Gamma \;=\; S^2\sigma T\cdot\frac{\phi(d_1)}{S\sigma\sqrt{T}} \;=\; S\phi(d_1)\sqrt{T} \;=\; \text{Vega}. \tag{20}$$

□ □

Intuition. This identity is the algebraic bridge between the two units in which a desk measures volatility exposure, and it says they are the same object up to a factor of σT. The *dollar gamma* ΓS^2 is the natural unit for *realized* volatility: it is the coefficient in the delta-hedged PnL, $\frac{1}{2}\,\Gamma S^2\big[(\Delta S/S)^2 - \sigma^2\,\Delta t\big]$, so it measures what the position earns or bleeds from the underlying actually moving. *Vega* is the natural unit for *implied* volatility: it measures how much the position is worth more or less when the market re-prices the surface. Rearranging the identity isolates the exchange rate between them,

$$\frac{\text{Vega}}{\Gamma S^2} \;=\; \sigma T,$$

which is proportional to time to maturity. This is the source of the standard trade-off between the two exposures. At a *fixed dollar gamma*, long-dated positions carry disproportionately *more* vega, because each unit of gamma is multiplied by a larger σT; short-dated positions are almost pure gamma and carry little vega. The same fact seen at a fixed position size runs through Vega $\approx 0.4\, S\sigma\sqrt{T}$ and $\Gamma \propto 1/\sqrt{T}$: shortening maturity raises gamma and lowers vega. Both readings are one statement viewed from the two ends of σT: gamma is the short-dated exposure, vega the long-dated one, and σT is what converts between them.

8. ATM theta and the $|\theta| \approx C/(2T)$ rule

8.1 Derivation

Result 8.1 (ATM theta)**.** *For an ATM European call with zero rates,*

$$\theta = \frac{\partial C}{\partial t} = -\frac{\partial C}{\partial T} \approx -\frac{S\sigma}{2\sqrt{2\pi T}}.$$

Proof. From the Brenner–Subrahmanyam approximation,

$$C_{\text{ATM}} \approx \frac{S\sigma}{\sqrt{2\pi}} \cdot \sqrt{T}. \tag{21}$$

Differentiating with respect to T:

$$\frac{\partial C}{\partial T} \approx \frac{S\sigma}{\sqrt{2\pi}} \cdot \frac{1}{2\sqrt{T}} = \frac{S\sigma}{2\sqrt{2\pi T}}. \tag{22}$$

Theta is the negative of this (the option loses value as time elapses, i.e., as T decreases):

$$\theta \approx -\frac{S\sigma}{2\sqrt{2\pi T}}. \tag{23}$$

□ □

8.2 Premium–theta ratio

For an ATM European option with zero rates:

$$\frac{|\theta|}{C} \approx \frac{1}{2T}.$$

Proof. Take the ratio of the previous two results:

$$\frac{|\theta|}{C} = \frac{S\sigma/(2\sqrt{2\pi T})}{S\sigma\sqrt{T}/\sqrt{2\pi}} = \frac{1}{2T}. \tag{24}$$

□ □

8.3 Daily theta in practical units

The theta above is per unit of calendar time measured in years. To convert to daily theta:

$$\theta_{\text{daily}} \approx \frac{|\theta|_{\text{annual}}}{365} = \frac{C}{2T_{\text{years}} \cdot 365}. \tag{25}$$

If the option has N calendar days to expiry, then $T = N/365$, and the 365 factors cancel:

$$\theta_{\text{daily}} \;\approx\; \frac{C}{2N}, \qquad N = \text{days to expiry.}$$

Days to expiry	Daily theta as fraction of premium
30 days	$C/60$
14 days	$C/28$
7 days	$C/14$
1 day	$C/2$

8.4 What the rule does and does not say

1. **Theta is not constant.** Since $|\theta| \propto 1/\sqrt{T}$ for ATM options, theta diverges as expiry approaches. The very large daily theta on near-expiry options is accompanied by very large gamma, which is precisely why short-dated short-vol positions accumulate risk so quickly.

2. **Theta scales with $\sqrt{T}$, not T.** Doubling time to expiry only multiplies daily theta by $\sqrt{2}$. Longer-dated options bleed less per day but more in total over their life.

3. **The ratio measures how fast premium decays right now.** The relation $|\theta|/C \approx 1/(2T)$ says that an ATM option loses a fraction $1/(2T)$ of its *current* value per unit time. A one-year ATM option bleeds about 1/2 of its value per year at today's rate; a one-month option ($T \approx 1/12$) bleeds about 6 times its value per year at today's rate—which is impossible to sustain, and is exactly the point: the rate is not constant. Because $|\theta| \propto 1/\sqrt{T}$, the decay accelerates as expiry approaches, so the premium is exhausted well before the naive extrapolation $C/|\theta| = 2T$ would suggest. The ratio is a snapshot of the instantaneous bleed, not a countdown.

4. **The approximation is ATM-only, and the deviation has a sign.** For *ITM* options much of the premium is intrinsic value, which does not decay, so $|\theta|/C < 1/(2T)$: they bleed proportionally *less* than the rule. For *OTM* options the premium is pure time value staked on an unlikely event and decays faster than $\sqrt{T}$ near expiry, so $|\theta|/C > 1/(2T)$: they bleed proportionally *more*. The ATM case, where $C \propto \sqrt{T}$ exactly to leading order, is the clean boundary between the two regimes.

9. The theta–gamma identity and the breakeven move

Section 6 connected vega and gamma. The second bridge—the one a desk lives by—connects theta and gamma, and unlike the ATM approximations it is *exact* at every strike.

Result 9.1 (Theta pays for gamma)**.** *For any European option with zero rates,*

$$\theta \;=\; -\tfrac{1}{2}\,\sigma^2 S^2\,\Gamma.$$

Proof. The Black–Scholes PDE with $r = 0$ reads $\theta + \frac{1}{2}\sigma^2 S^2\Gamma = 0$; rearranging gives the identity. Alternatively, direct substitution of the formulas already derived:

$$-\tfrac{1}{2}\,\sigma^2 S^2 \cdot \frac{\phi(d_1)}{S\sigma\sqrt{T}} \;=\; -\,\frac{S\,\sigma\,\phi(d_1)}{2\sqrt{T}} \;=\; \theta.$$

□

Interpretation. Theta is the *rent* paid for holding gamma. A delta-hedged long option position loses $|\theta|\,dt$ with certainty as time passes, and earns $\frac{1}{2}\Gamma\,(dS)^2$ from realized movement. The identity says these two cash flows are priced to cancel *when the underlying realizes exactly the implied volatility.* Long gamma is a bet that realized movement will exceed what was paid; short gamma collects the rent and owes the movement.

9.1 The daily breakeven move

Setting the gamma gain equal to the theta bleed over a step dt,

$$\tfrac{1}{2}\,\Gamma\,(dS)^2 \;=\; \tfrac{1}{2}\,\sigma^2 S^2\,\Gamma\,dt \quad\Longrightarrow\quad \left|\frac{dS}{S}\right| \;=\; \sigma\sqrt{dt}.$$

Gamma cancels: the breakeven move is the same for every option. Over one trading day ($dt = 1/252$),

$$\left|\frac{dS}{S}\right|_{\text{daily}} \;=\; \frac{\sigma}{\sqrt{252}} \;\approx\; \frac{\sigma}{16}.$$

The rule of 16. A 16-vol underlying must move about 1% a day for a delta-hedged long option to break even; a 32-vol underlying, about 2%. Annualized volatility divided by 16 is daily volatility, because $\sqrt{252} \approx 15.87$. This is the most-used mental conversion on any options desk, and it is nothing but the theta–gamma identity read at a one-day horizon: the implied vol *is* the daily breakeven, restated in annual units.

10. Summary card

Quantity	**Formula at ATM**	**Source**
ATM call premium	$0.4\,S\sigma\sqrt{T}$	Brenner–Subrahmanyam
ATM straddle premium	$0.8\,S\sigma\sqrt{T}$	$C + P = 2C$ at ATM
Vega (per $\Delta\sigma = 1.0$)	$0.4\,S\sqrt{T}$	$S\phi(d_1)\sqrt{T}$
Vega (per vol point)	$0.004\,S\sqrt{T}$	Vega $\div\,100$
Gamma	$\dfrac{0.4}{S\sigma\sqrt{T}}$	$\phi(d_1)/(S\sigma\sqrt{T})$
Vega–gamma identity	Vega $= S^2\sigma T\,\Gamma$	Algebraic
Theta (annualized)	$-\dfrac{0.4\,S\sigma}{2\sqrt{T}}$	$-\partial C/\partial T$
Daily theta (premium ratio)	$\dfrac{C}{2N}$, $N =$ days to expiry	$\lvert\theta\rvert/C = 1/(2T)$

The unifying observation. Every constant 0.4 in this table is $1/\sqrt{2\pi} = \phi(0)$, the value of the standard normal density at its peak. It appears in every formula because, for an at-the-money short-dated option, $d_1 \approx 0$, and the ATM Greeks are all proportional to $\phi(d_1)$. One geometric truth — the height of the normal at zero — generates the entire table.

A. $\sqrt{T}$ at a glance

Every formula in these notes is linear in $\sqrt{T}$, so fast mental pricing reduces to knowing $\sqrt{T}$ for the standard horizons. Maturities use month fractions of a year; the one-day row uses trading time (1/252), which is the convention for daily volatility scaling.

Horizon	T (years)	$\sqrt{T}$	ATM call, $\sigma = 20\%$ ($0.4\,\sigma\sqrt{T}$, % of spot)
1 trading day	1/252	$0.063 \approx 1/16$	0.5%
1 week	1/52	0.139	1.1%
2 weeks	2/52	0.196	1.6%
1 month	1/12	0.289	2.3%
2 months	1/6	0.408	3.3%
3 months	1/4	**0.500**	4.0%
6 months	1/2	0.707	5.7%
9 months	3/4	0.866	6.9%
1 year	1	1.000	8.0%

Three anchors carry the whole table. **Three months is one half**: $\sqrt{1/4} = 0.5$ exactly, so a 3-month ATM option costs $0.2\,\sigma$ of spot—the cleanest anchor in mental pricing (on 20 vol: 4%). **One trading day is one sixteenth**: $\sqrt{1/252} \approx 1/16$, the rule of 16. **Doubling maturity multiplies by** $\sqrt{2} \approx 1.4$: premium, vega and the breakeven move all scale the same way, which is why a 6-month option costs only 41% more than a 3-month one, not twice as much. The last column doubles as a sanity check: quote it for $\sigma = 20\%$ and rescale linearly in σ for anything else.

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